

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: CodeSparks

Project / Product Name: Fixflow - Smart Guided Troubleshooting Engine (Samsung PRISM GenAI Hackathon, Theme 2)

Organization / Institution (if any): SRM Institute of Science and Technology (SRMIST)

Submission Date: 30 September 2026

Team Members: B. Parkavi, Aishwarya S Prakash

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes

AI tools were used selectively during development for assistance with implementation, testing, documentation, and refinement. The team reviewed and validated the final work.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: AI assistants were used to discuss architecture, semantic-cache strategies, retrieval improvements, API design, evaluation strategy, and troubleshooting workflow. Final design decisions were made by the team.

Code generation or assistance: AI coding assistance was used for Python/FastAPI implementation, retrieval, semantic-cache logic, extraction, deeplink matching, API work, tests, and debugging. Changes were reviewed and tested by the team.

UI / UX design: AI assistance was used where applicable for frontend/UI planning and implementation. The final interface and user flow were selected and refined by the team.

Content creation: AI assistance was used to draft and refine technical documentation and presentation material. Content was checked against the implemented system.

Data analysis: AI assistance was used to inspect benchmark outputs, cache-hit/miss patterns, latency, retrieval behavior, and regression results. Reported measurements come from the team's own benchmark runs.

Testing / debugging: AI coding assistance helped identify bugs, propose regression tests, and debug failures. The team verified changes through the automated test suite and benchmark runs.

4. Feature Origin Classification

Feature 1

1. Feature Name: Hybrid SIIS Retrieval
2. Self-Generated/AI-Generated/Both: Both
3. Description: Combines TF-IDF, BM25, device affinity, and semantic/context reranking to select relevant SIIS scenarios. Retrieval weights and symptom handling were tuned by the team.

Feature 2

1. Feature Name: Two-tier Semantic Cache
2. Self-Generated/AI-Generated/Both: Both
3. Description: Uses L1 normalized exact matching and L2 semantic matching with scenario metadata, symptom families, and device/model compatibility. Candidate handling and compatibility logic were tuned from benchmark results.

Feature 3

1. Feature Name: Deterministic SIIS Extraction
2. Self-Generated/AI-Generated/Both: Both
3. Description: Converts SIIS titles and procedure headings into structured goals, actions, step groups, and categories without inventing source steps. Regression fixes were implemented and tested by the team.

Feature 4

1. Feature Name: Semantic Deeplink Matching
2. Self-Generated/AI-Generated/Both: Both
3. Description: Matches action/step evidence against the supplied masked Galaxy Settings deeplink catalog. Matching and fallback behavior were implemented and regression-tested.

Feature 5

1. Feature Name: Response Validation and API Contract
2. Self-Generated/AI-Generated/Both: Both
3. Description: Produces and validates the ContextDeeplinkResponse structure, deeplink fields, formatting constraints, and URL-leakage rules. API behavior was verified through tests.

Feature 6

1. Feature Name: Benchmark and Regression Suite
2. Self-Generated/AI-Generated/Both: Both
3. Description: Provides repeatable benchmark diagnostics for cache hit rate, latency, coverage, schema validity, and URL leakage, with regression tests selected and interpreted by the team.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. Yes

No proprietary or copyrighted data misused. I Agree

6. Declaration & Sign-Off

Name of Team Representative: B. Parkavi

Role: Team Lead

Signature: B. Parkavi

Date: 30 September 2026